A PHYSICALLY DISAGGREGATED COMPUTING SYSTEM

Reinaldo Javier Aguiar Marcano

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Title: A PHYSICALLY DISAGGREGATED COMPUTING SYSTEM

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A computing system architecture where primary computer components are physically disaggregated into separate enclosures and interconnected by an ultra-high-bandwidth, low-latency data fabric. The system comprises a first chassis containing a plurality of processors, a second chassis containing a plurality of memory modules, and potentially other chassis for storage and accelerator cards. The data fabric provides communication between the disaggregated components at speeds sufficient to allow them to operate as a single, cohesive logical computer, effectively "stretching" the motherboard across a data center. This architecture overcomes the thermal, power, and resource-stranding inefficiencies of traditional monolithic server designs.

Background of the Invention

The design of modern computers and servers is monolithic, with all primary components—CPU, memory, storage, and GPUs—tethered to a single motherboard within a single chassis. This design is a direct consequence of the distance and bandwidth limitations of electrical interconnects like PCIe and DDR memory buses. This physical coupling leads to significant inefficiencies in power, cooling, and resource allocation, as the entire chassis must be provisioned for the worst-case component, and resources cannot be scaled independently. This "tyranny of the motherboard" creates inflexible and inefficient data center infrastructure.

Summary of the Invention

The present invention is a physically disaggregated computing system that decouples the primary components of a computer into independent, fungible resource pools. The system fundamentally redefines a computer from a static, monolithic box to a dynamic, logically composed entity.

The system comprises at least two physically distinct enclosures or chassis: a first containing a pool of processors, and a second containing a pool of memory modules. Additional enclosures may house pools of storage or accelerator resources. The key enabling component is an ultra-high-bandwidth data fabric, capable of multi-terabit-per-second data rates, that interconnects these separate enclosures. The fabric's bandwidth is so high and its latency so low that it functions as a distributed, chassis-to-chassis backplane, allowing a processor in one chassis to access memory in another chassis with performance comparable to a traditional motherboard. This allows for the dynamic composition of "bare-metal" computers, where a specific set of resources is logically assembled for a task and then disassembled, returning the components to the pools.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting a traditional monolithic server architecture with the physically disaggregated system of the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, the system is contrasted with prior art. A traditional monolithic server (100) contains all components—CPU (101), Memory (102), and Storage (103)—physically coupled on a single motherboard (104) within a single chassis. The disaggregated computing system of the present invention (110) separates these components into physically distinct, specialized chassis. A first chassis (111) contains a pool of CPUs, a second chassis (112) contains a pool of memory, and a third chassis (113) contains a pool of storage. These separate chassis are interconnected by an ultra-high-bandwidth optical data fabric (114), which allows the disparate resources to function as a single, logically composed computer.

Claims

What is claimed is:

1. A computing system, comprising:

    - a first chassis containing one or more processors;
    - a second chassis, physically separate from the first chassis, containing one or more memory modules; and
    - a high-bandwidth data fabric providing a communicative link between at least one processor in the first chassis and at least one memory module in the second chassis, the fabric having a data transfer rate sufficient to allow the processor to access the memory module as main memory.

2. The system of claim 1, further comprising a third chassis containing one or more storage devices, communicatively coupled to the data fabric.

3. The system of claim 1, further comprising a fourth chassis containing one or more accelerator cards, communicatively coupled to the data fabric.

4. The system of claim 1, wherein the data fabric has a data transfer rate exceeding 10 terabits per second.

5. The system of claim 1, wherein the first chassis and the second chassis are optimized with different cooling and power delivery systems.

6. The system of claim 1, wherein the high-bandwidth data fabric is a massively parallel optical data fabric.

7. The system of claim 6, wherein the fabric uses optical interconnects as described in U.S. Patent Application No. 19/559,705.

Drawings

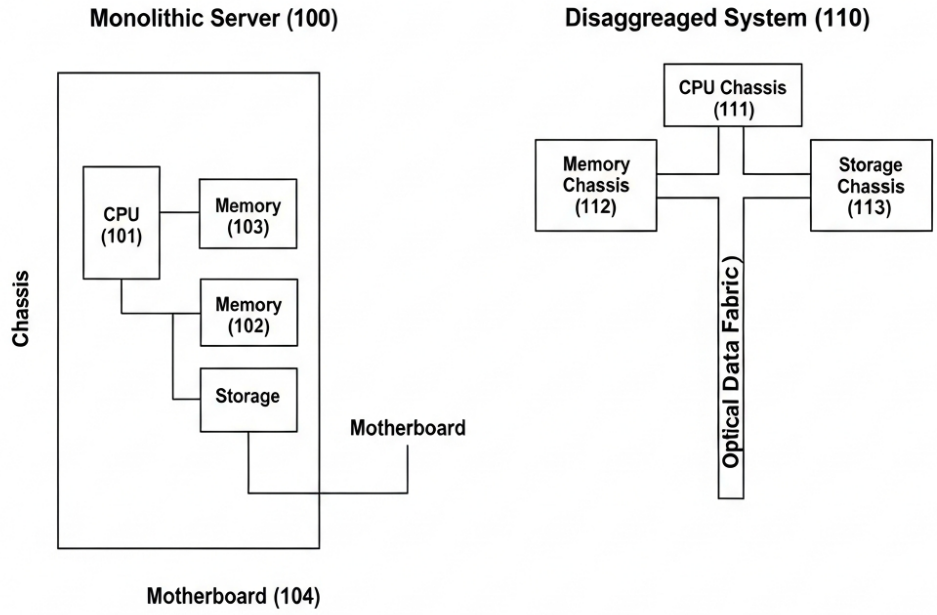


Figure 1: Conceptual diagram contrasting a traditional monolithic server architecture with the physically disaggregated system of the present invention.